Product Requirements Document

GBM2804H-100T Basic Motor Rotation

Overview

Develop firmware for the NUCLEO-G431RB with the X-NUCLEO-IHM16M1 to demonstrate basic rotation of the GBM2804H-100T three-phase gimbal motor. The purpose of this project is only to verify the MCU, motor-driver board, PWM outputs and motor connections. Closed-loop FOC and motor characterization are intentionally excluded.

Target Hardware

Item	Specification
MCU Board	NUCLEO-G431RB
Motor Driver	X-NUCLEO-IHM16M1 / STSPIN830
Motor	GBM2804H-100T, 7 pole pairs
Supply	12 V DC
Motor Outputs	U, V, W three-phase output

Functional Requirements

1. Configure TIM1 to generate three-phase PWM.
2. Use the factory-default X-NUCLEO-IHM16M1 three-input configuration.
3. Generate a rotating three-phase electrical field using open-loop sinusoidal PWM.
4. Start with low PWM amplitude and gradually ramp the electrical frequency.
5. Motor shall remain stopped after system initialization.
6. Support clockwise and counter-clockwise rotation.
7. MOTOR START shall begin the open-loop startup ramp.
8. MOTOR STOP shall immediately disable PWM.
9. An active STSPIN830 EN/FAULT condition shall immediately disable PWM.
10. STBY shall be controlled correctly to enable and disable the motor driver.

Initial Operating Parameters

Parameter	Initial Value
PWM switching frequency	20 kHz
Initial PWM amplitude	10%
Maximum PWM amplitude for this test	20%
Initial electrical frequency	1 Hz
Electrical-frequency ramp target	10 Hz
Motor pole pairs	7

No closed-loop RPM regulation is required. Electrical frequency is only used to create a rotating field; actual motor RPM is not measured or guaranteed.

UART Interface

Command	Action

MOTOR START	Start the open-loop rotation ramp
MOTOR STOP	Disable PWM and stop the motor
DIR CW / DIR CCW	Set rotation direction
FREQ <Hz>	Set electrical frequency
DUTY <%>	Set PWM amplitude within the configured safe limit
STATUS	Report motor state, direction, electrical frequency, PWM duty and driver fault status

Validation Criteria

- Firmware builds and flashes successfully.
- PWM remains disabled after boot.
- MOTOR START produces visible motor movement or rotation.
- MOTOR STOP disables PWM and stops motor drive.
- CW and CCW commands reverse the generated phase sequence.
- Driver fault disables PWM.

Explicitly Out of Scope

Sensorless FOC, Motor resistance, inductance or flux characterization, Rotor observer, Clarke/Park and inverse transforms, Current PI or speed PI controller, Closed-loop RPM control, Phase-current ADC control, VBUS ADC control, ADC-triggered DMA, Current-offset calibration, Oscilloscope integration, Automatic motor-parameter identification.

Implementation constraint: Do not block MOTOR START because ADC measurements, motor electrical parameters, characterization, FOC, observer, current sensing, or closed-loop control are unavailable. These are explicitly outside the scope of this project. MOTOR START shall depend only on successful PWM/GPIO initialization and the absence of an active STSPIN830 hardware fault.